

Huntington's disease is caused by a dominant allele of the gene that codes for the production of the huntingtin protein. This protein affects the development of many different tissues, including brain tissue.

- The Huntington allele contains several repeats of the base sequence CAG, which codes for glutamine.
- This results in a polyglutamine section in the synthesised protein.
- A gene with more than 39 CAG repeats produces a protein that does not fold properly and does not function.
- Symptoms of Huntington's disease usually first appear between the ages of 30 and 45 years.
- There is no treatment for the disease, which is progressive and always fatal.
- Some people with between 27 and 35 CAG repeats do not develop the disease, but may still pass on the Huntington allele to their children, who may develop the disease as the number of repeats tends to increase when gametes are produced.

(a) (i) The amino acid sequences on either side of the polyglutamine section of the huntingtin protein are not changed by the presence of the CAG repeats in the Huntington allele.

Explain why this is so.

.....[2]

(ii) With reference to the information given, explain why Huntington's disease cannot be treated with gene therapy.

.....[4]

(b) Young people who have a parent with Huntington's disease can choose to be screened for the presence of the Huntington allele.

(i) State the probability that a young person who has **one** parent with Huntington's disease will inherit the Huntington allele.

.....[1]

(ii) Suggest **one** advantage and **one** disadvantage of screening for Huntington's disease before any symptoms occur.

advantage

.....

.....

disadvantage

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.....[2]

(c) A couple, in which one partner has the Huntington allele, may choose to use IVF (*in vitro* fertilisation) to have a child.

Any embryos obtained from the IVF procedure can be screened in the following way:

- carry out an embryo biopsy
- use PCR
- test for the presence of the Huntington allele
- only implant embryos that do not contain the Huntington allele.

(i) State what is meant by the term *embryo biopsy*.

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(ii) Explain why PCR is used in this procedure.

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.....[1]

(iii) Outline **two** social or ethical implications of screening embryos in this way.

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[Total: 13]